



Society of Petroleum Engineers

SPE-229209-MS

Revolutionizing Water Cut Monitoring: Strategic Field Deployment of a Data Science Framework

Khalid Al-Dohaiem, Bibi Hussain Akbar, and Bogi Haryo Nugroho, Innovation & Digitalization Team, Kuwait Oil Company, Kuwait City, Kuwait; Brahantiyo Aji Sumarto, NFD, BA, North Kuwait, Kuwait Oil Company, Kuwait City, Kuwait; Joel Titus, SLB Kuwait, Kuwait City, Kuwait

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229209-MS](https://doi.org/10.2118/229209-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper details the design and phased implementation of an automated, data-driven water cut estimation system across 150 wells in Kuwait Oil Company's NFD (BA) - NK field, addressing the limitations of conventional PGOR-based workflows that relied on infrequent well tests and manual analysis. These legacy processes delayed water breakthrough detection and constrained proactive reservoir management. The objective was to deliver a real-time, scalable, and highly accurate solution aligned with KOC's Digital Transformation Strategy to minimize oil losses and accelerate decision-making.

The system was developed using the OSEMN data science framework within the Dataiku platform, leveraging Python-based workflows for data ingestion, cleansing, and modeling. ESP sensor readings, PGOR test data, and well metadata were integrated through automated pipelines connected to KOC's Oracle database. Rigorous preprocessing addressed missing values, outliers via Z-score thresholds, and normalization. Exploratory analysis employed K-Means clustering to segment wells by water cut behavior, enabling tailored model development. Predictive models were built using linear regression with hyperparameter tuning and Grid search optimization. Real-time dashboards and automated email alerts were deployed to provide engineers with actionable insights and early warnings of abnormal water cut trends.

Field deployment reduced water cut evaluation time from 2.5 months to less than two hours for all 150 wells—a 5,400× improvement—while achieving median R^2 values near 0.7 across varying water cut regimes. Beta testing on 40 wells validated system stability and predictive accuracy prior to full rollout. The solution prevented an estimated 1.29 million barrels of production loss, valued at approximately USD 77.2 million, by enabling earlier intervention. Key challenges included inconsistent ESP data and variable field conditions, mitigated through fallback routines and continuous model refinement.

This work represents one of the first large-scale applications of integrated data science and real-time analytics for water cut monitoring in brownfield operations. The approach establishes a replicable model for digital oilfield initiatives, demonstrating how advanced analytics can transform production surveillance and enhance reservoir management efficiency.

Background & Industry Challenge

Timely and accurate water cut (WC) estimation is fundamental to operational excellence and optimal realization of economic value from oil production. The traditional method adopted by the KOC was based on spreadsheets and manually compiled information from different locations. The conventional method, depending on periodic PGOR tests and manual processing, was error-prone and not scalable to give an overall real-time snapshot of the field. Delay in processing data for planning and taking preventive action resulted in preventable production losses. Depending on infrequent or late water cut data, an unexpected rise in water cut would significantly affect oil production and economic feasibility. Classical PGOR tests, although useful for single-point measurement, were not equipped with continuous monitoring necessary for real-time, proactive decision-making and production optimization

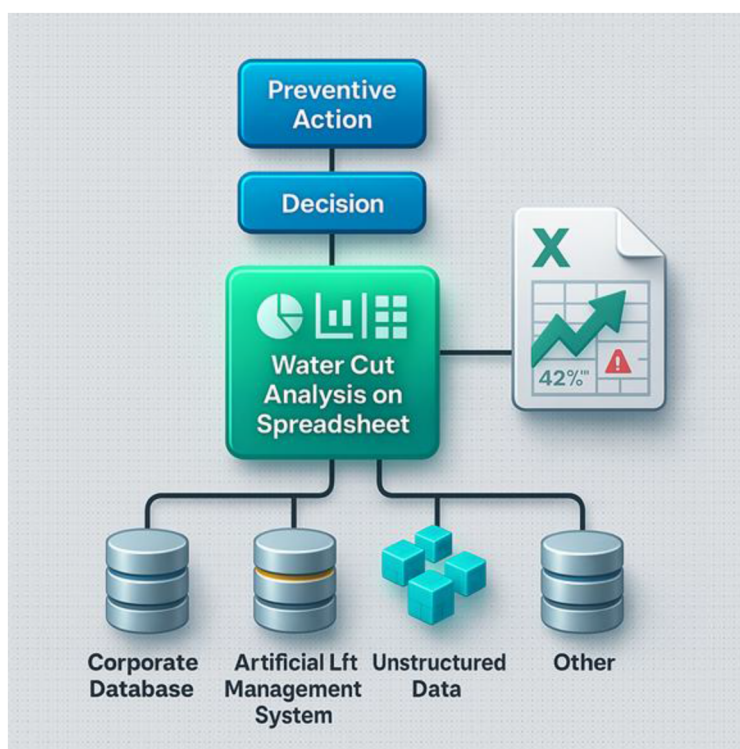


Figure 1—Conventional Water Cut Analysis Workflow

The strategic need was to shift from this reactive, intuition-based approach toward a proactive, real-time surveillance system powered by data science and artificial intelligence. The goal was to develop a scalable and automated solution that could integrate data from various sources, generate insightful visualizations, and enable data-driven decisions. The journey of moving from a conventional, spreadsheet-based workflow to a data science-driven, real-time monitoring system redefined surveillance speed at scale.

Objectives

The primary objective of this paper is to present the development and phased rollout of a smart, data-driven system designed to accurately estimate and predict water cut in real time. The project aimed to replace traditional PGOR-based workflows, often slow and manual, with an automated solution that accelerates decision-making, reduces oil losses, and supports Kuwait Oil Company's broader Digital Transformation Strategy. To achieve this, the team rebuild and enhanced the existing water cut estimation framework, rigorously tested its performance under real operating conditions, and successfully implemented the optimized solution across 150 wells in the NFD (BA) - NK field. The location of the field is shown

in Figure 2. The approach followed the OSEMN data science model—Obtain, Scrub, Explore, Model, and iNterpret, to structure and streamline the entire process.

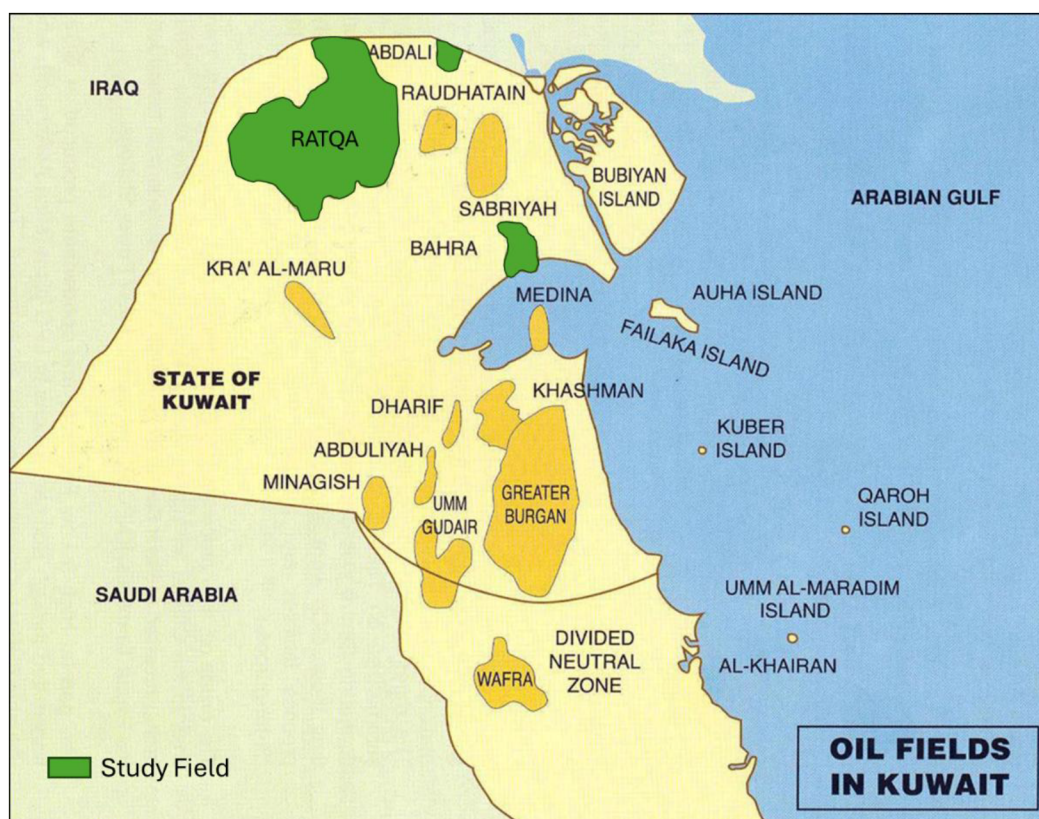


Figure 2—Geographic field map or schematic showing the deployment area in KOC's North Kuwait field.

The key goals of the project were:

- Implement the enhanced water cut estimation solution across targeted 150 wells in the NFD (BA) - NK field. The
- Transform water cut monitoring into a real-time capability by leveraging continuous ESP sensor readings and PGOR test data.
- Improve engineering productivity and decision-making through automation and advanced analytics.
- Ensure the solution aligns with KOC's digital transformation objectives and operational excellence targets.

The scope of work extended beyond a simple rollout. It involved refining the OSEMN-based framework for greater accuracy and stability, validating the system through rigorous field-based testing, and ensuring seamless integration into existing operational workflows. The solution was engineered to utilize multiple critical data sources, ESP sensor readings and depth, PGOR test data, and deviation surveys, each contributing unique insights to strengthen model robustness. Key activities included:

- Enhanced Implementation: Functional and component-level improvements to optimize predictive performance.
- Rigorous Testing: Comprehensive validation under real operating conditions, benchmarked against traditional workflows and previous pilots.

- Phased rollout: Coordinated deploy across targeted wells, supported by active engagement with end-users to ensure adoption and sustainability.

This structured approach ensured the final solution was not only technically optimized but also operationally impactful, enabling faster, data-driven decisions and setting a strong foundation for scalable digital transformation in mature fields.

Technical Methodology: Data Science Pipeline

The solution was developed and deployed using the OSEMN data science framework (Obtain, Scrub, Explore, Model, and iNterpret), implemented with Python within the Dataiku environment. Dataiku offered an integrated setting that allowed easy, streamlined workflow, intuitive teamwork, and robust Python support, all of which were needed to effectively manage an enterprise-scale project.

OSEMN Framework Breakdown

The project followed the main steps of the OSEMN framework:

- Obtain: Ingestion pipelines were created to retrieve ESP sensor values and PGOR test data from KOC's Oracle database.
- Scrub: This critical phase involved rigorous data cleaning and preprocessing. It included handling missing values, outlier removal using Z-score thresholds, and normalization to prepare the datasets for analysis.
- Explore: Exploratory data analysis was performed to identify trends, patterns, and data clusters. A key innovation was the use of K-Means dynamic clustering to segment water cut regimes for customized modeling.
- Model: Regression models were trained with regression algorithms optimized with hyperparameter tuning. A special Python script was utilized for automatic optimization and model creation according to pre-defined criteria.
- iNterpret: This final step involved translating model outputs into practical, actionable insights through real-time dashboards and automated email alerts.

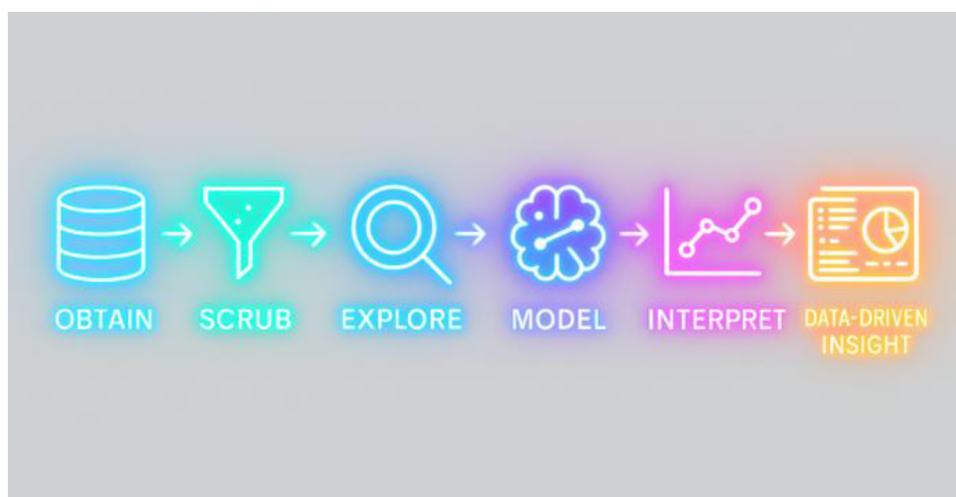


Figure 3—Visual flow diagram of the OSEMN process from data acquisition to insight

Model Design & Optimization

The models are linear regression variant models that customized to predict water cut from ESP sensor data and PGOR well test data. The model performance was evaluated through R-squared (R^2) and Mean Absolute

Error (MAE) on different water cut zones. The model building involved cross-validation and cluster-specific tuning to create robust models. The system involved a user-preference model that allowed engineers to tune Z-score ranges and weighting factors, combining computational power with human judgment for enhanced predictability.

Table 1—Comparison of Model Performance by Model Type and Water Cut Regime.

Model Type	Water Cut Regime	R-squared (R2)	Mean Absolute Error (MAE)
Linear Regression	Low	High (>0.7), with peak at 1.0	Low
Linear Regression	Medium	Moderate (Median ~0.5)	Moderate
Linear Regression	High	High (>0.7), with peak at 1.0	Low

System Architecture

System design is an integrated loop with Dataiku as the central platform for modeling and data processing. Data from the Oracle database is ingested, wrangled through Dataiku pipelines, and utilized to train and execute predictive models. The output is sent to a user interface for visualization and an automated email service for alerting. The system was architected to be robust, with fallback routines and data validation rules for dealing with missing or incomplete inputs and being resilient. A central element of the architecture is the real-time loop, which facilitates monitoring and daily updates.



Figure 4—System Architecture showing the data flow from data sources to the final user interface and alerts

Deployment & Field Integration

The deployment was a phased approach, building upon the success of earlier project phases. The project successfully transitioned from a beta phase with 40 wells to a full deployment across all 150 wells in the NFD (BA) - NK field.

- Beta Phase: The solution underwent rigorous beta testing across 40 wells to confirm its prediction accuracy and system stability. This phase featured a full, end-to-end demonstration of the solution's workflow, from raw data extraction to live predictions and reporting.

- Rollout: Following successful beta testing, the enhanced solution was deployed across 150 wells. This rollout was carefully orchestrated to integrate smoothly with current operations and systems, with comprehensive training provided to field engineers.

The deployment results showed consistent accuracy R-squared (R^2) values 0.7 – 1 across low and high water cut all 150 wells, proving the reliability of the machine learning models in live operations. The new user interface and real-time visuals made the system easier and faster for operations teams to use. The automated notifications and reports continued to function dependably, enabling prompt responses to potential problems.

The solution demonstrates a revolutionary transformation in water cut monitoring delivered through a robust dashboard and smart alert system, as illustrated in Figures 5 and 6. Figure 5 presents a comprehensive Water Cut Monitoring Dashboard, designed to visualize and analyze water cut data across multiple dimensions. The dashboard integrates spatial, time-based, and statistical perspectives, making it a powerful tool for monitoring, diagnosing, and optimizing water cut performance in oilfield operations. This represents a transformative shift from reactive to a continuous, preventive monitoring approach, enabling proactive decision-making and enhanced reservoir management.

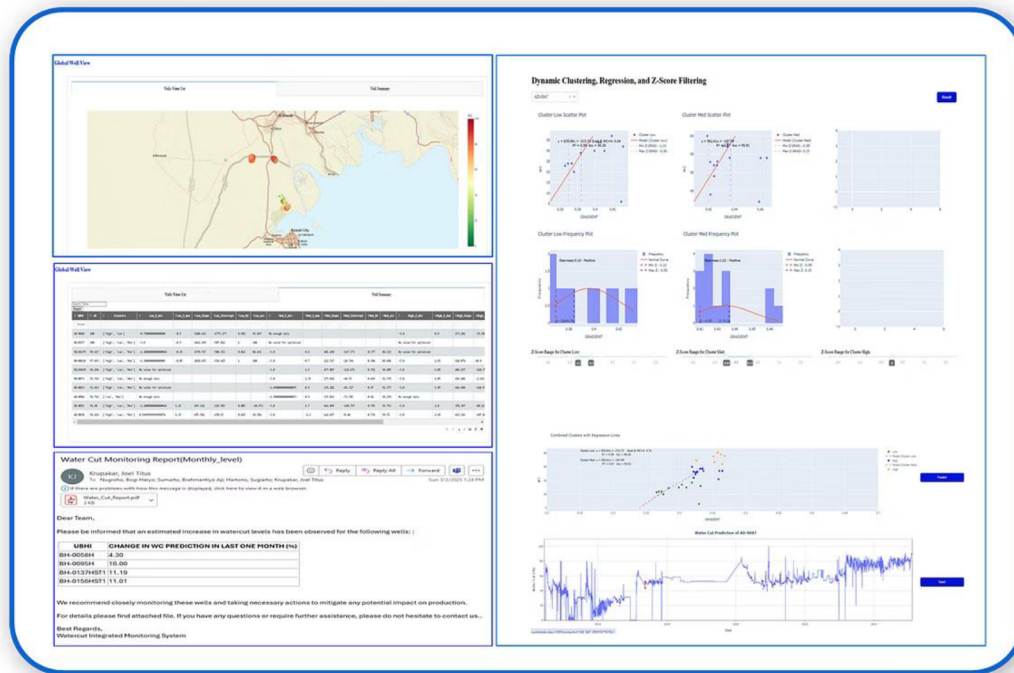


Figure 5—Water Cut Monitoring Dashboard screenshot.

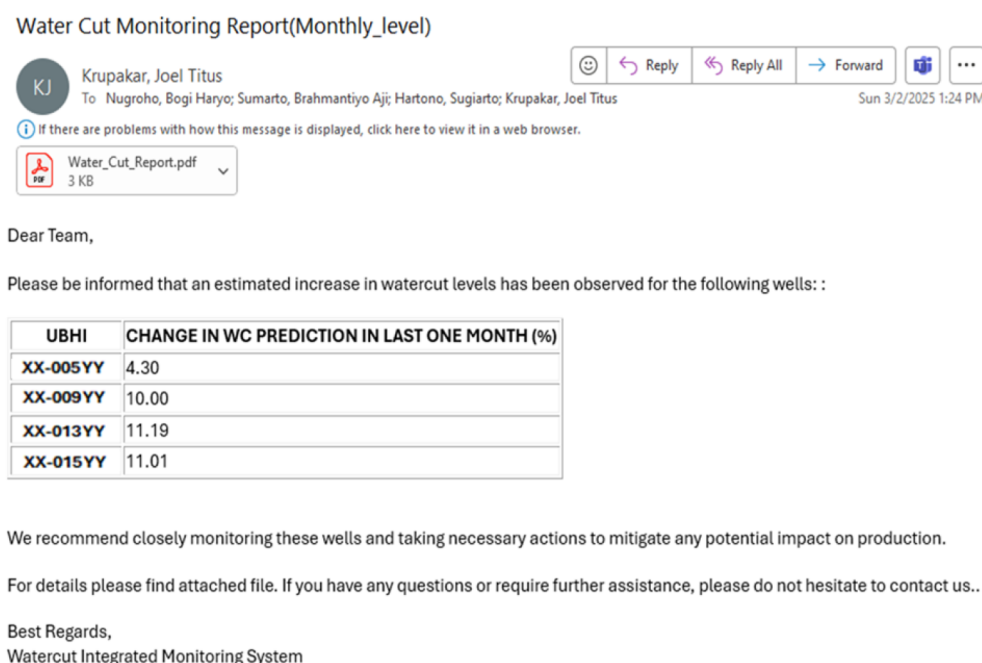


Figure 6—Water Cut Prediction Email Alert notifying engineers of estimated increases in water cut levels.

Figure 6 displays an automated email notification sent to field engineers, highlighting wells with rising water cut levels. and urges immediate monitoring and mitigation to protect production efficiency. The concise format ensures timely awareness and action.

Performance Results & Value Unlocked

The implementation of the data science framework for water cut monitoring revealed substantial value and tangible benefits for KOC.

Performance Metrics

- **Analysis Speed:** The time required for an operational water cut analysis was dramatically reduced from over 2.5 months to under two hours for 150 wells, a 5,400x speedup factor. The system consistently averaged about one minute per well.
- **Predictive Accuracy:** The models demonstrated reliable performance with median R-squared (R²) values 0.7 - 1 across low and high water cut zones. The performance of the models varied by regime, with low and high water cut models showing stronger performance than the medium water cut model.

Business Value & Operational Impact

Implementation prevented an estimated 1.29 million barrels of lost oil production, valued approximately USD 77.2 million, based on a value of \$60 per barrel. This tremendous value was realized through enabling early water breakthrough detection and allowing engineers to perform proactive, preventively motivated actions.

Furthermore, the system provided:

- **Rapid Decision-Making:** Moving towards real-time data and automated alarms enabled engineers to respond to changes in production within minutes, not months, and bettered engineer decisions and quicker response.
- **Improve Production Optimization:** The system assists groups in crafting data-driven approaches to enhance production performance, optimize oil loss, and make better use of ESP sensor information.

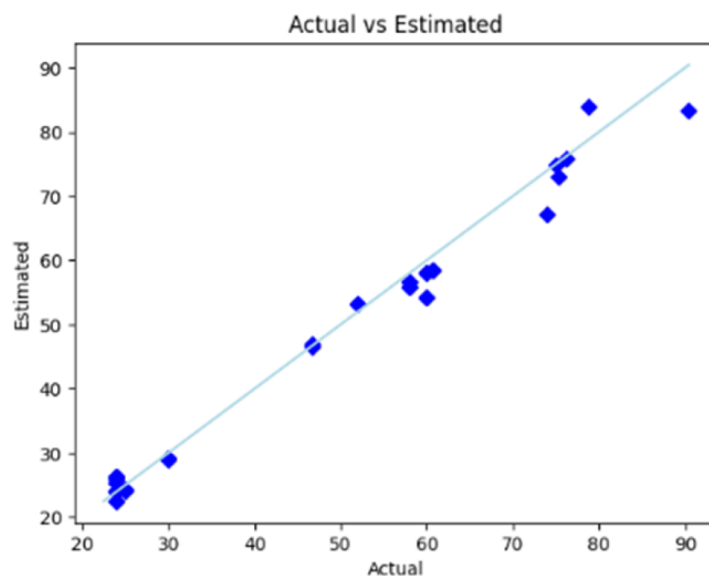


Figure 7—Plot: actual vs. predicted WC Well XX-0007

Challenges & Lessons Learned

The project encountered several challenges, providing valuable lessons for future initiatives.

- Variation in Data Quality: There were condition of ESP signal losses, lost readings, and PGOR data sources in the system. One of the challenges was that some variables, like GRADIENT, had skewed data distributions, which can affect the accuracy of linear regression models.
 - Mitigation: Fallback rules and resilient reasoning were implemented to address inference gaps. It was also highlighted that real-time data quality monitoring, filtering data sources and model retraining approaches are essential for accommodating changing field conditions.
- Modeling Complex Regimes: The model's performance was less consistent in the medium water cut regime ($R^2 \sim 0.5$), likely due to transitional flow characteristics or operational complexity.
 - Mitigation: A key lesson learned was the need for specific models for different regimes, reservoir-based models to get more reliable predictions. It was recommended to explore reservoir based, non-linear models and other advanced machine learning techniques to capture complex relationships.
- Adoption Curve: The project had to overcome user skepticism to build operational trust in the new system.
 - Mitigation: Utilization of a beta period with engineer input and feedback loops, in combination with clear visualization and alerting, assisted in user adoption.

Table 2—Technical Challenges and Mitigation Strategies.

Technical Challenge	Mitigation Strategy
Data quality variability (missing ESP data, incorrect PGOR entries)	Fallback rules, resilient logic, and automated data validation to handle inference gaps.
Modeling complexity of medium water cut regimes	Explore non-linear models and other advanced machine learning techniques; build specific models for different regimes.
Infrastructure limitations (non-GPU servers)	Recommend a transition to GPU-powered servers for live deployment to accelerate processing and ensure scalability.

Innovation Value & Broader Industry Contribution

This project marks one of the first large-scale implementations of an integrated, data science-based water cut monitoring system within brownfield operations. The solution provides a scalable, repeatable model that can be implemented in additional mature fields, offshore, injector, or gas facilities.

The innovative value lies in the seamless integration of several key elements within a single, unified workflow:

- **Automated ESP Analytics:** The system utilizes real-time, continuous ESP sensor readings to give dynamic, real-time information traditional methods were unable to provide.
- **Clustering Methods:** K-Means dynamic clustering was utilized to classify data and create personalized models for various regimes of flow, enhancing predictive performance and facilitating easy data analysis. Exploartion data analysis to maximize the model performance.
- **Real-Time Visualization and Alerting:** The project went a step further from mere data processing through building of intuitive dashboards and automated alerting, which gave engineers the information required for proactive management.

This paper shows a replicable, scalable framework that combines machine learning, real-time visualization, and alert-based workflow. This strategy establishes a new standard for proactive reservoir management and encourages innovation and enhanced competitive advantage across the petroleum industry.

Table 3—Comparative table: This solution vs. legacy tools vs. global benchmark.

Feature	Legacy Tools (Spreadsheets)	This Solution (Dataiku/Python)	Global Benchmark
Analysis Speed	Very Slow (2.5 months)	Very Fast (under 2 hours)	Varies, but moving towards real-time
Data Integration	Manual, fragmented	Automated, unified from Oracle DB	Automated, multi-source integration
Scalability	Low, manual overhead	High (proven on 150 wells)	High, cloud-based solutions
Monitoring	Infrequent, reactive	Continuous, proactive with alerts	Continuous, AI-driven alerts
Cost	Low direct cost, high hidden cost of lost production	Higher software cost, significant savings from loss prevention	Varies, high ROI from efficiency gains

Conclusion & Forward Path

This rollout has delivered more than just a technical solution, it is a working model of how data science can transform field operations. By applying the OSEMN framework within the Dataiku platform, the system now supports data-driven decision-making with remarkable speed and accuracy. Key achievements include:

- Strong gradient-water cut relationships and high predictive accuracy ($R^2 > 0.7$ across low and high water cut regimes).
- Analysis speed has increased 5,400 times—turning hours into seconds.
- Operational impact is real: an estimated 1.29 million barrels of oil saved, worth \$77.2 million.

This success has led to the system being widely accepted as a reliable approach for field surveillance and setting a clear, proven model that can be replicated across the upstream sector.

To further strengthen and scale the solution, the following enhancements are proposed:

- *Data Quality & Modeling:* Implement real-time data quality monitoring and frequent model updates, especially for medium water cut regimes with high variability. Explore non-linear and advanced machine learning models to improve accuracy.

- *Technological Upgrades*: Integrate high-speed GPU processing to support faster and more insightful visualizations.
- *Feature Expansion*: Extend the solution to support production optimization, reservoir modeling, and ESP analytics, aligning with broader digital transformation and operational excellence goals.

This effort lays a strong foundation for continuous monitoring and intelligent field management. With targeted enhancements, the solution stands as a replicable model for the entire E&P industry, driving innovation, efficiency, and smarter decision-making.

Acknowledgments

Innovation & Technology Group, NFD (BA) - NK Team, Innovation & Digitalization Team, advisory members, and contributors recognized in this project for guidance and collaboration enabling this phased deployment.

References

1. Akbar, B. H., Al-Aradi, H. J., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. *Harnessing Data Science for Enhanced Water Cut Analysis in NK-NFD*. Kuwait University Scientific Poster Day, Kuwait, February, 2025.
2. Nugroho, B. H., Akbar, B. H., Sumarto, B. A., Al-Aradi, H. J., Joenaedy, M. A., & Achmad, P. R. *Harnessing Data Science for Produced Water Estimation: Unleashing the Power of Data-driven Techniques*. SPE-219016-MS, SPE Water Lifecycle Management Conference & Exhibition, Abu Dhabi, March, 2024.
3. Akbar, B. H., Al-Aradi, H. J., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. (2024, December). *Accelerating Insights Automated Water Cut Estimation for Enhancing Production Optimization*. KPC and Google Cloud event, Kuwait.
4. Al-Aradi, H. J., Nugroho, B. H., Akbar, B. H., Sumarto, B. A., Joenaedy, M. A., & Achmad, P. R. *Data Science Driven Decision Making Insight into Water Ratio Production Level*. Middle East Water Week Conference and Exhibition, Al Khobar-KSA., December, 2023.
5. Sumarto, B. A., Nugroho, B. H., Akbar, B. H., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Water Cut Estimation Using ESP Sensor Reading and Well Test Data, A Novel Approach*. SPE-211407-MS, ADIPEC, Abu Dhabi. November, 2022.
6. Akbar, B. H., Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Data Science For Water Cut Analysis System Utilizing ESP Sensor Data Prototype, Pilot and Its Potential Impact*. SPE-211406-MS, ADIPEC, Abu Dhabi, November, 2022
7. Nugroho, B. H., Sumarto, B. A., Joenaedy, M. A., Al-Aradi, H. J., & Achmad, P. R. *Data Science for Water Cut Analysis System Utilizing ESP Sensor Data, Conceptual Model, and Its Proposed Solution*. SPE-207624-MS, ADIPEC, November, 2022.
8. Nugroho, B. H., Sumarto, B. A., Akbar, B. H., Titus, J. *Data Science for Water Cut Estimation in NFD (BA) - NK Phase 4 – Field Deployment Project Report*. KOC, Kuwait, March, 2025.
9. Nugroho, B. H., & Achmad, P. R. *Data Science for Water Cut Estimation in NFD (BA) - NK Phase 2 & 3 - Data Quality Control, Prototyping & Pilot Project Report*. KOC, Kuwait, March, 2021.
10. Nugroho, B. H., & Achmad, P. R. *Data science for Water Cut Estimation in NFD (BA) - NK - Phase 1 Assessment Project Report*. KOC, Kuwait, January, 2021.
11. Sumarto, B. A. *ESP General Operation and Technical Procedures*. Personal proprietary and limited internal disclosure, Kuwait, 2019.

12. Eric, J., *Data Science Life Cycle: CRISP-DM and OSEMN frameworks*, Website, *Data Science Life Cycle: CRISP-DM and OSEMN frameworks*.
13. Data Science PM, *OSEMN Data Science Life Cycle*, Website, OSEMN Data Science Life Cycle Data Science PM, March, 2024.
14. Damlapinar, M. *Analytics of Life: Making Sense of Data Analytics, Machine Learning and the Artificial Intelligence*. NLITX, New York, 2019.
15. Hastie, T., Tibshirani, R., & Friedman, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer, New York, 2009.
16. Rahman, R. *From Information to Impact: How Data Drives Decision-Making*. DEV Community, October, 2024.
17. Liashchynskyi, P., & Liashchynskyi, P. *Grid Search, Random Search, Genetic Algorithm: A Big Comparison for NAS*, Cornell University, New York, 2019.
18. Bergstra, J., & Bengio, Y. *Random Search for Hyper-Parameter Optimization*, *Journal of Machine Learning Research* **13**, p: 281–305, 2012
19. Mujahid Tabassum, Kuruvilla Mathew., A Genetic Algorithm Analysis towards Optimization Solutions, *International Journal of Digital Information and Wireless Communications (IJDIWC)*, **4** (1), p: 124–142, January, 2024.
20. Institute of Data. *How Businesses Can Best Use Data Science to Support Operations*, April, 2024.
21. *The data science venn diagram*: <http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram>, March, 2013.
22. *Microsoft Excel Limitation*: <https://www.ibm.com/support/knowledgecenter/en/SSD29G2.0.0/com.ibm.swg.ba.cognos.tmlinst.2.0.0.doc/cexcellimits.html>.